

Ismail Aissaoui
Email: i.aissaoui@esi-sba.dz
Phone: +213 660 70 77 96
Portfolio: ismail-aissaoui.vercel.app

G rald Rocher (Univ C te d'Azur)

Subject: Application for PhD Position – Distributed Abductive Reasoning for Self-Explaining Digital Twins

Dear G rald Rocher,

I am applying for the PhD position on "Distributed Abductive Reasoning for Self-Explaining Digital Twins" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

A core focus of my engineering work has been Explainable AI (XAI) and autonomous systems. While engineering the Autonomous Cyber-Shield Platform, I utilized Reinforcement Learning and Adversarial Co-Evolution to secure networks, but the critical challenge was understanding the 'why' behind the agent's decisions. I am eager to apply my background in deep learning to develop distributed abductive reasoning frameworks that allow digital twins to not just predict states, but logically explain their systemic inferences.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui